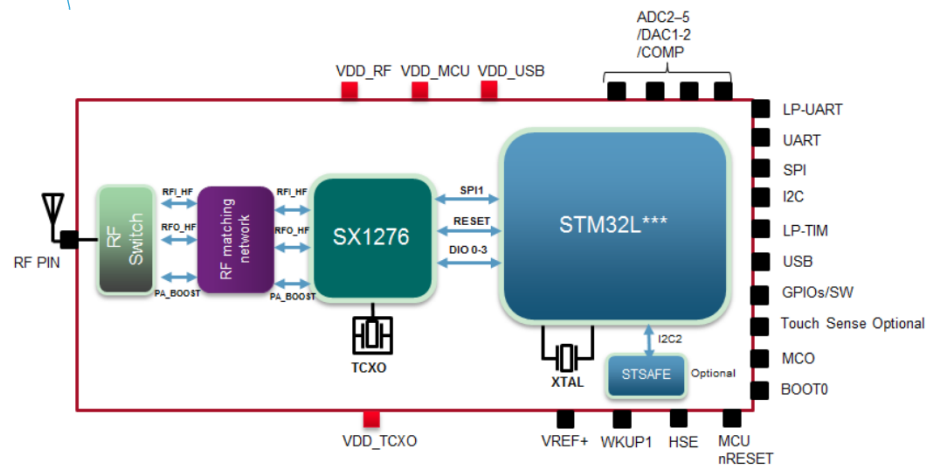


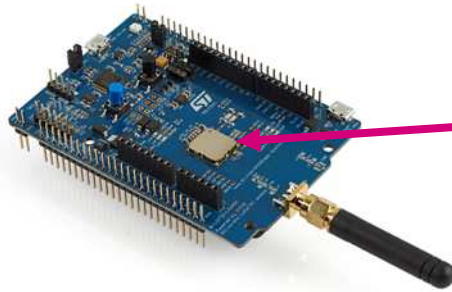


STM32L0

What is hidden under metal cover ?

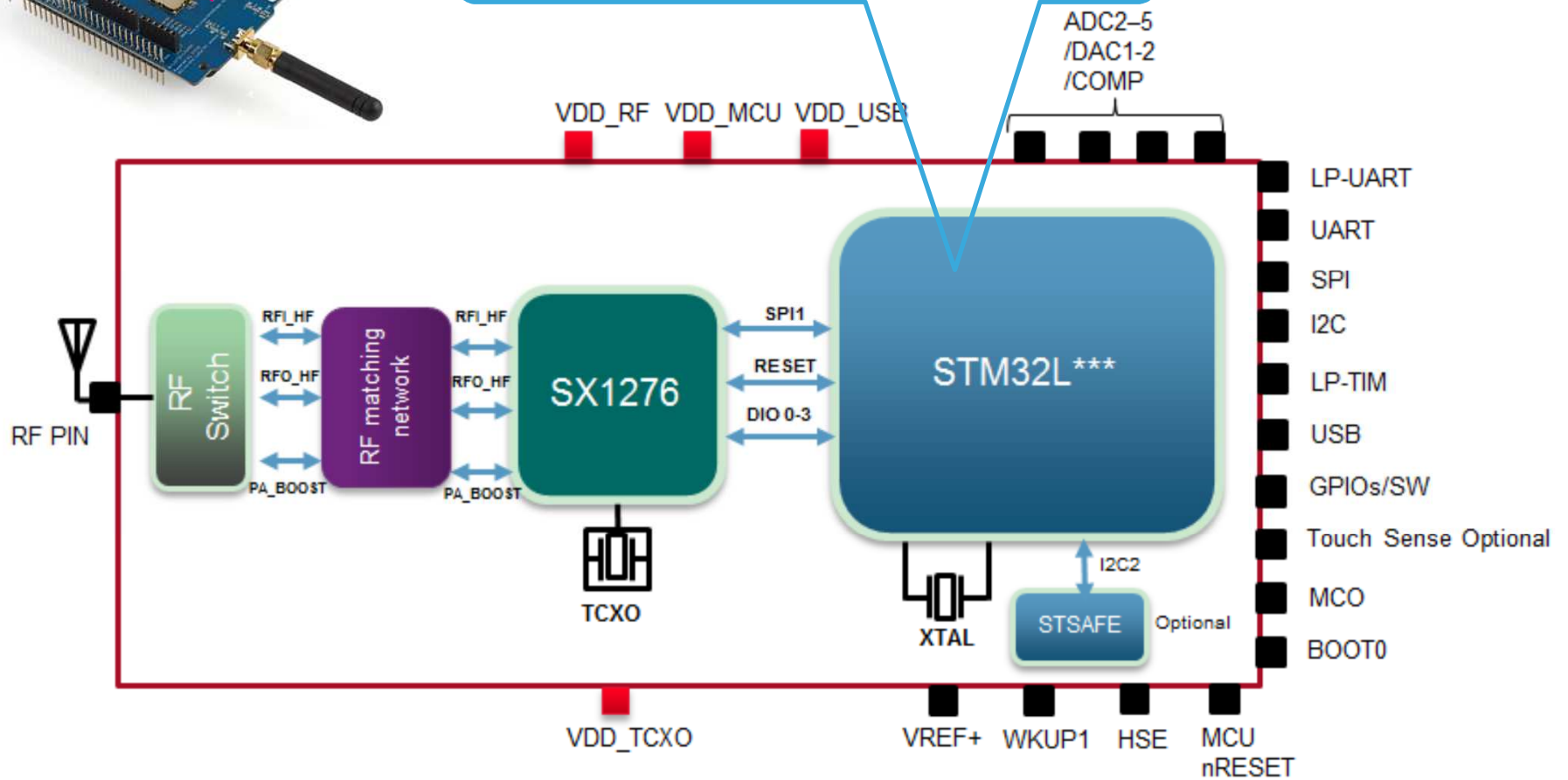


Murata module



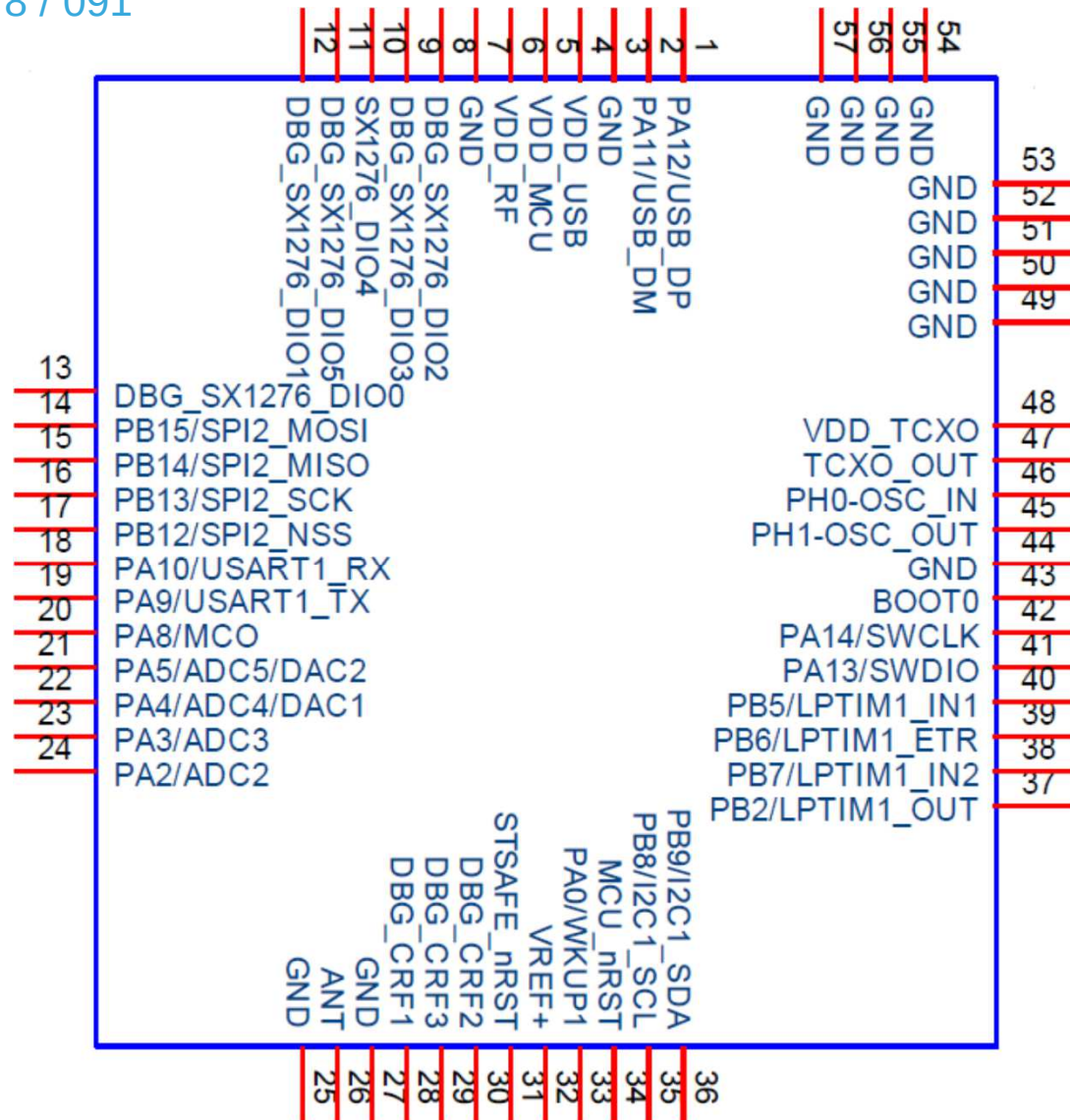
STM32L082CZ: Murata CMWX1ZZABZ-078

STM32L072CZ: Murata CMWX1ZZABZ-091



Pinout of
CMWX1ZZABZ-078 / 091

Murata module



DS

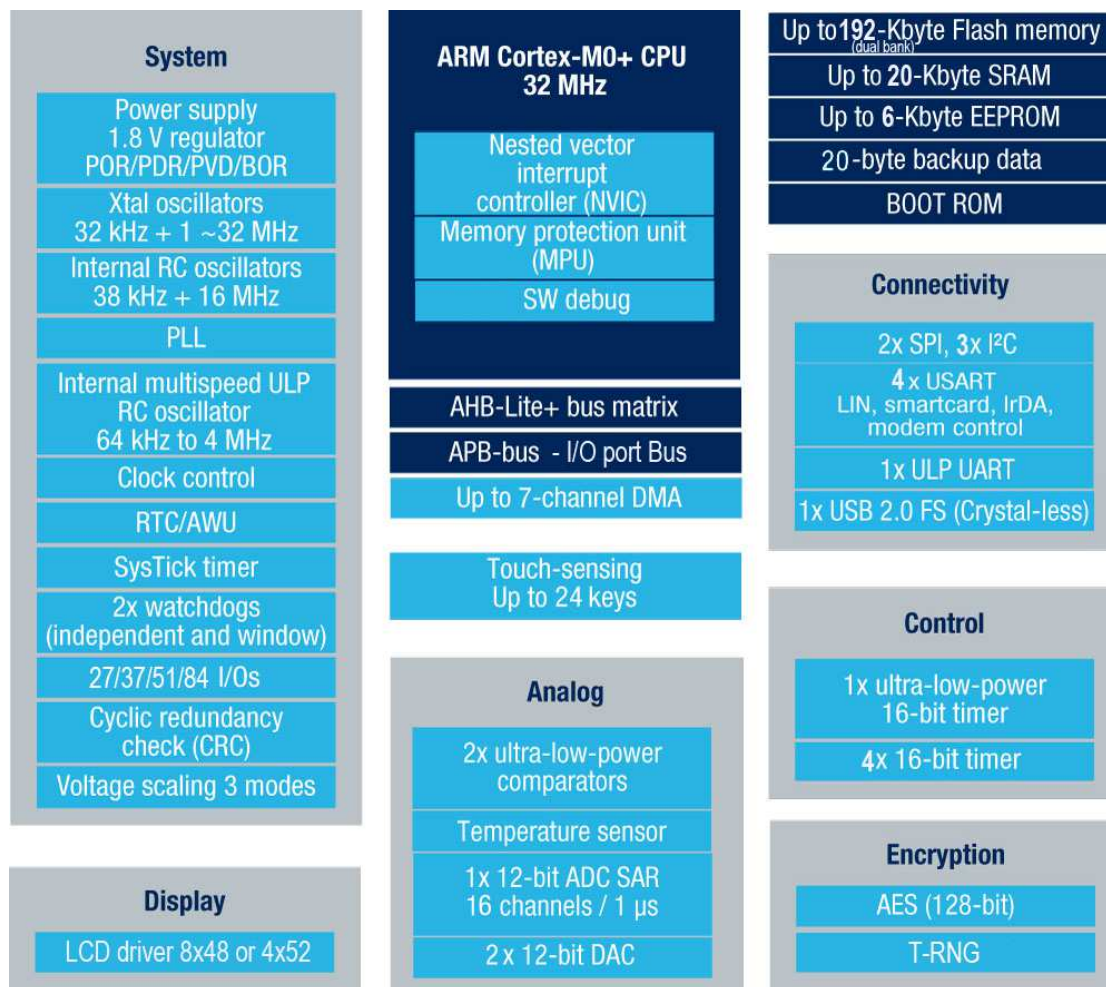
http://wireless.murata.com/RFM/data/type_abz.pdf



STM32L08x block diagram

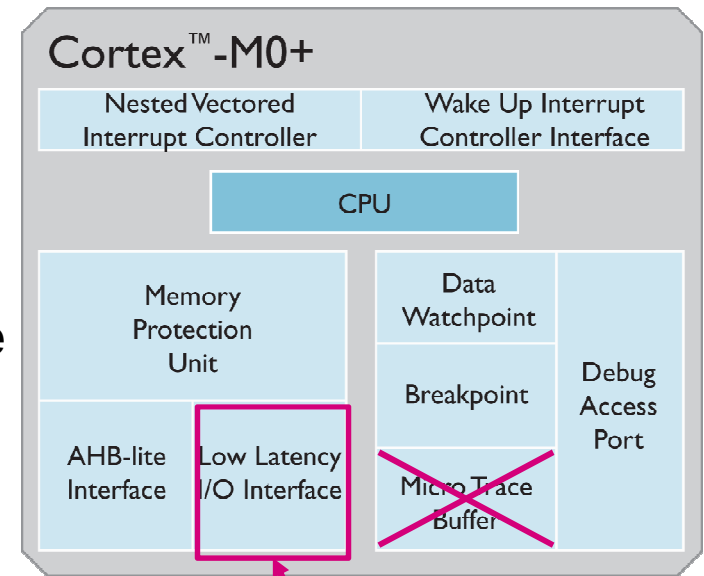
- Key features

- ARM Cortex-M0+ at 32MHz
 - Single-cycle I/O access
 - Single-cycle multiplier (MUL)
 - 0.95 DMIPS/MHz
- 1.71V to 3.6V, 32MHz full functional
- Digital down to 1.65V
- -40°C to +125°C temperature range
- ADC with build-in HW oversampling
 - Down to 1.65V
- Flash + Ram code sector lock
- USB 2,0 FS certified
 - Build-in 48MHz oscillator
 - Battery Charger Detection
 - Link Power Management
- Independent clock domain
 - I2C, USART/UART
 - USB
- 5x timers
 - 1x 16-bit (4ch)
 - 3x 16-bit(2ch)
 - 1x 16-bit LP¹ available in stop



Cortex-M0+ Processor Overview

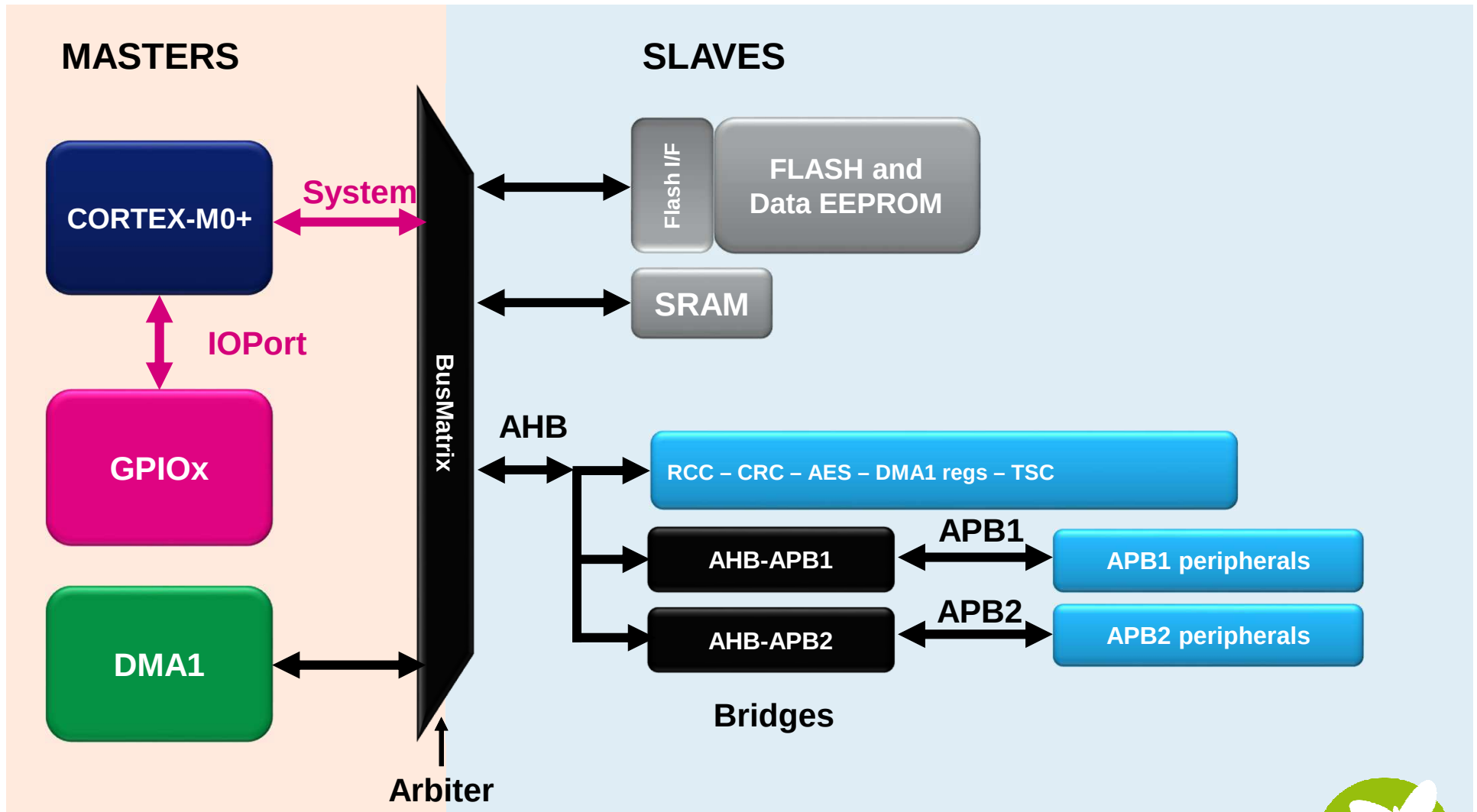
- **ARMv6-M** Architecture
- **von Neumann** architecture, **2-stage pipeline**
- **Integrated** Nested Vectored Interrupt Controller (**NVIC**) for low latency interrupt processing
- Designed to be fully programmed in **C-language**
- Vector Table is a simple list of addresses
- Full **Thumb** Instruction Set and **subset of Thumb-2**
- **Single cycle multiply (optional)**
- **Memory Protection Unit (MPU)* (optional)** , **privileged / unprivileged mode***
- Integrated 24-bit System Timer (**SysTick**) for RTOS (optional)



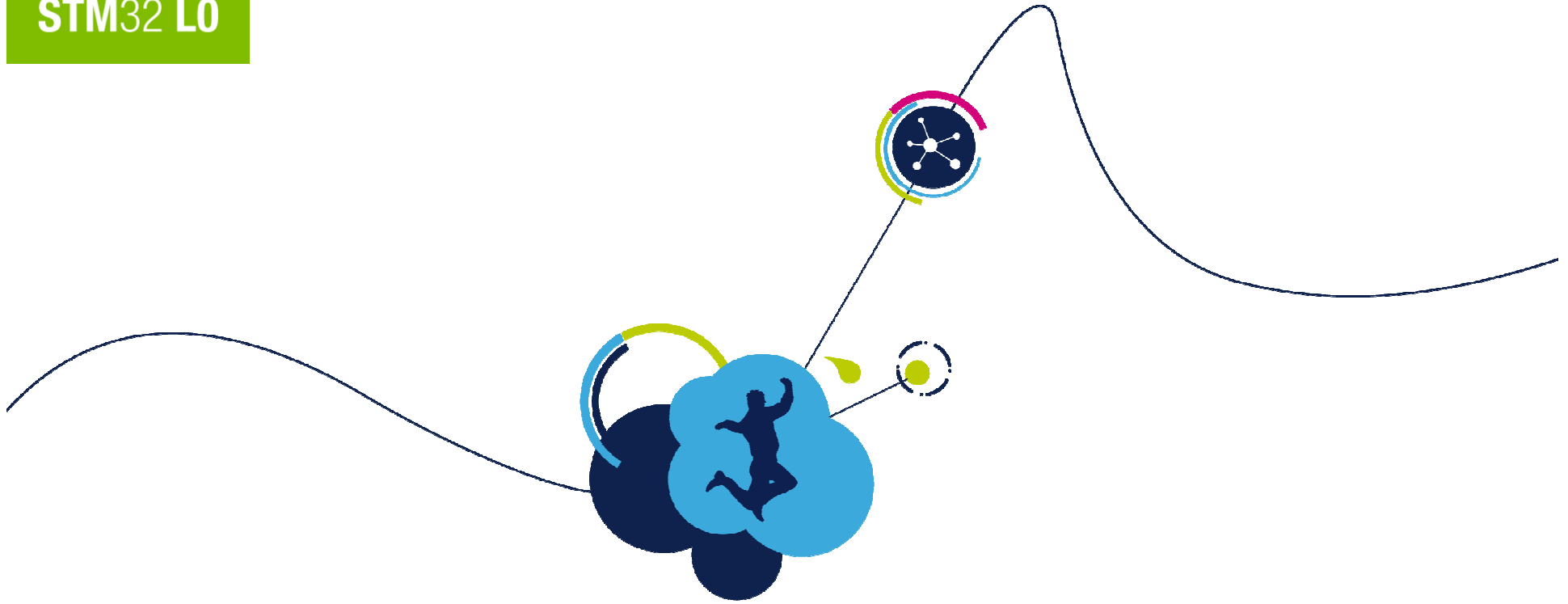
IOPort

+ interrupt vector table relocation

System Architecture



STM32 L0

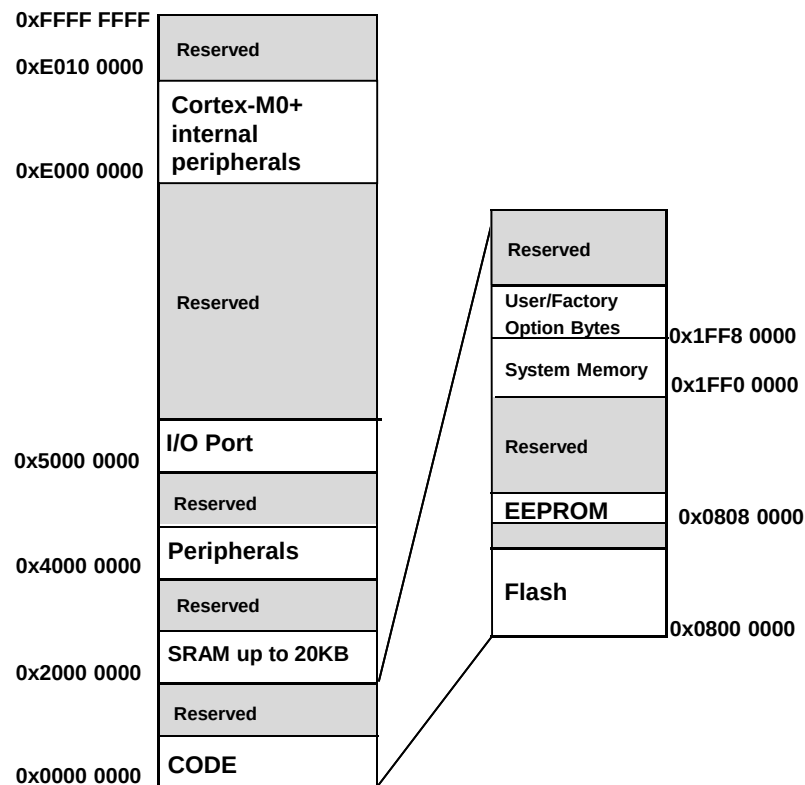


System blocks

Internal Memories

Memory Mapping and Boot Modes

- Addressable memory space of 4 GB
- RAM : up to 20 Kbytes
- FLASH : up to 192 Kbytes
- Data EEPROM: up to 6 Kbytes



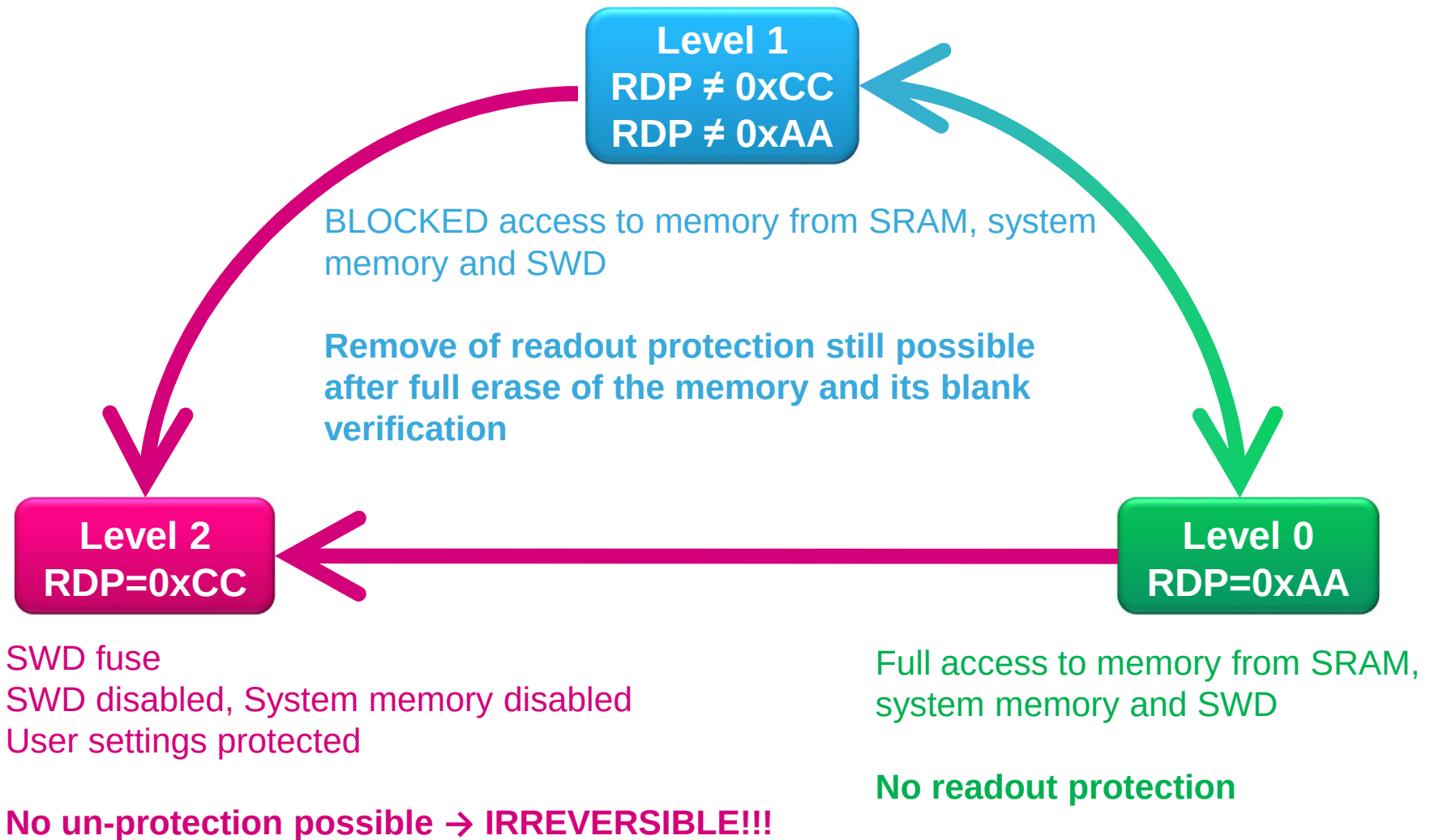
Boot modes

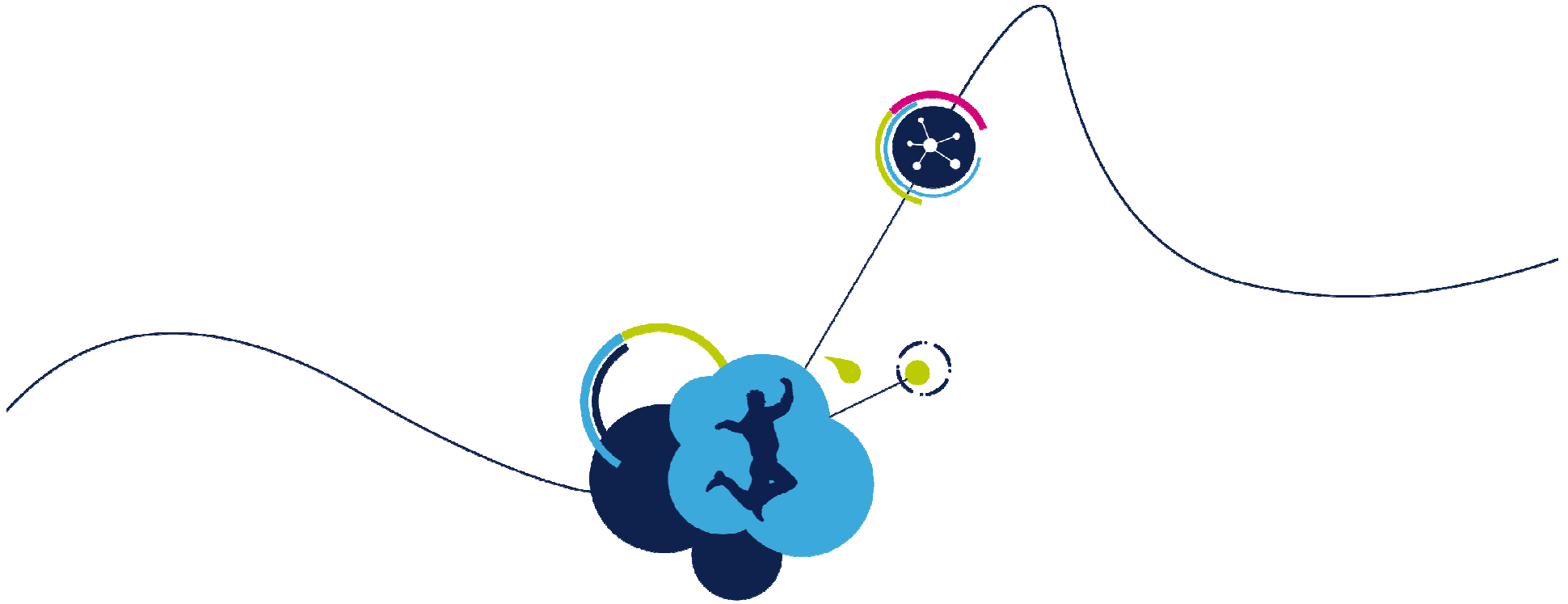
Depending on the Boot configuration, Embedded Flash Memory, System Memory or Embedded SRAM Memory is aliased at @0x00. The System Memory and Embedded SRAM memory can be remapped also at @0x0 using a dedicated software bits. BOOT0 is read on dedicated pin BOOT0. BOOT1 is an option register bit.

BOOT Mode Selection		Boot Mode	Aliasing
nBOOT1	BOOT0		
x	0	User Flash	User Flash is selected as boot space
0	1	SystemMemory	SystemMemory is selected as boot space
1	1	Embedded SRAM	Embedded SRAM is selected as boot space

- **SystemMemory:** contains the Bootloader used to re-program the FLASH through USART1/2 or SPI1/2
- **Boot from SRAM** In the application initialization code you have to Relocate the Vector Table in SRAM using the NVIC Exception Table and Offset register.

Flash Read Protection (FLASH)



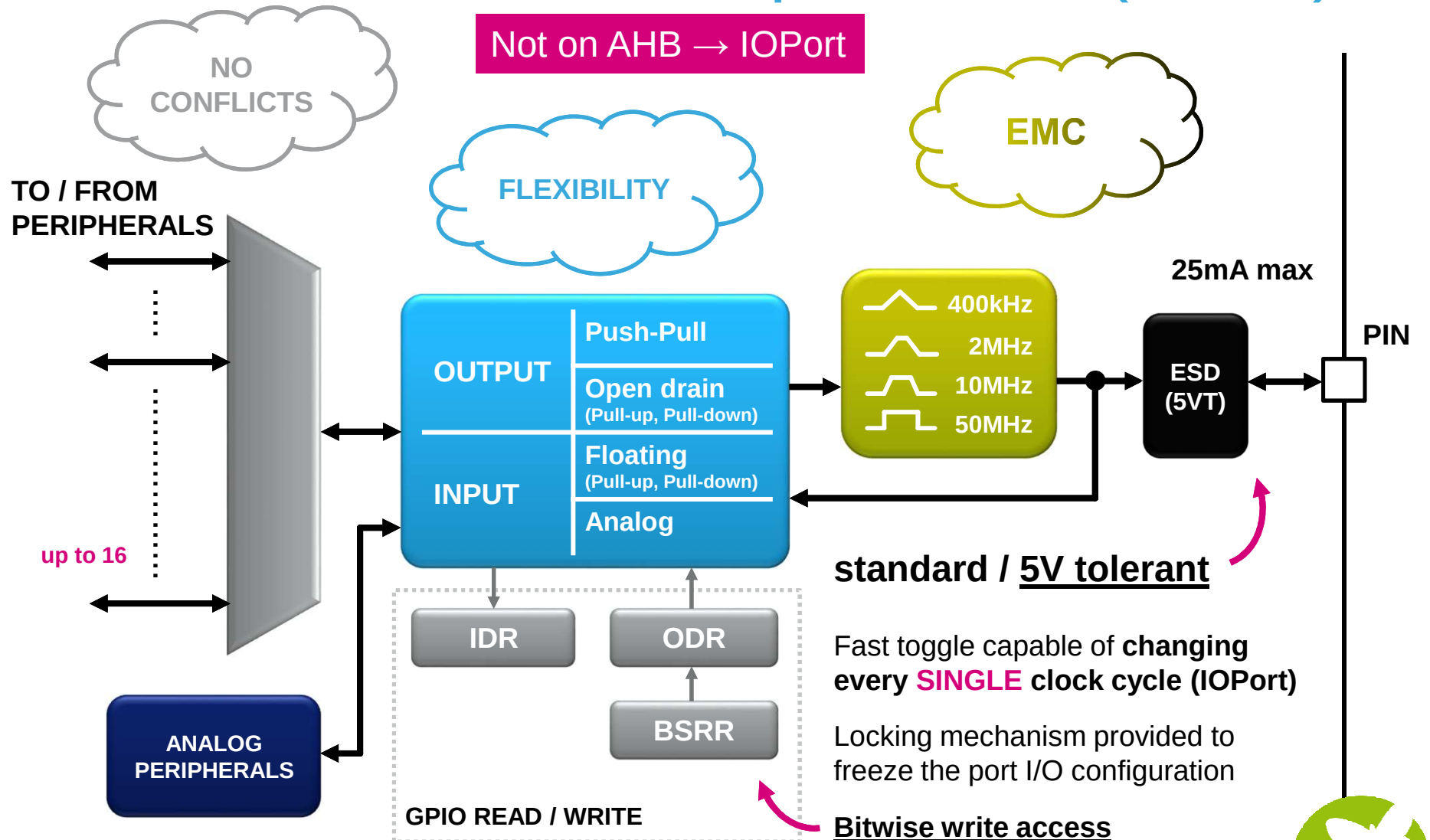


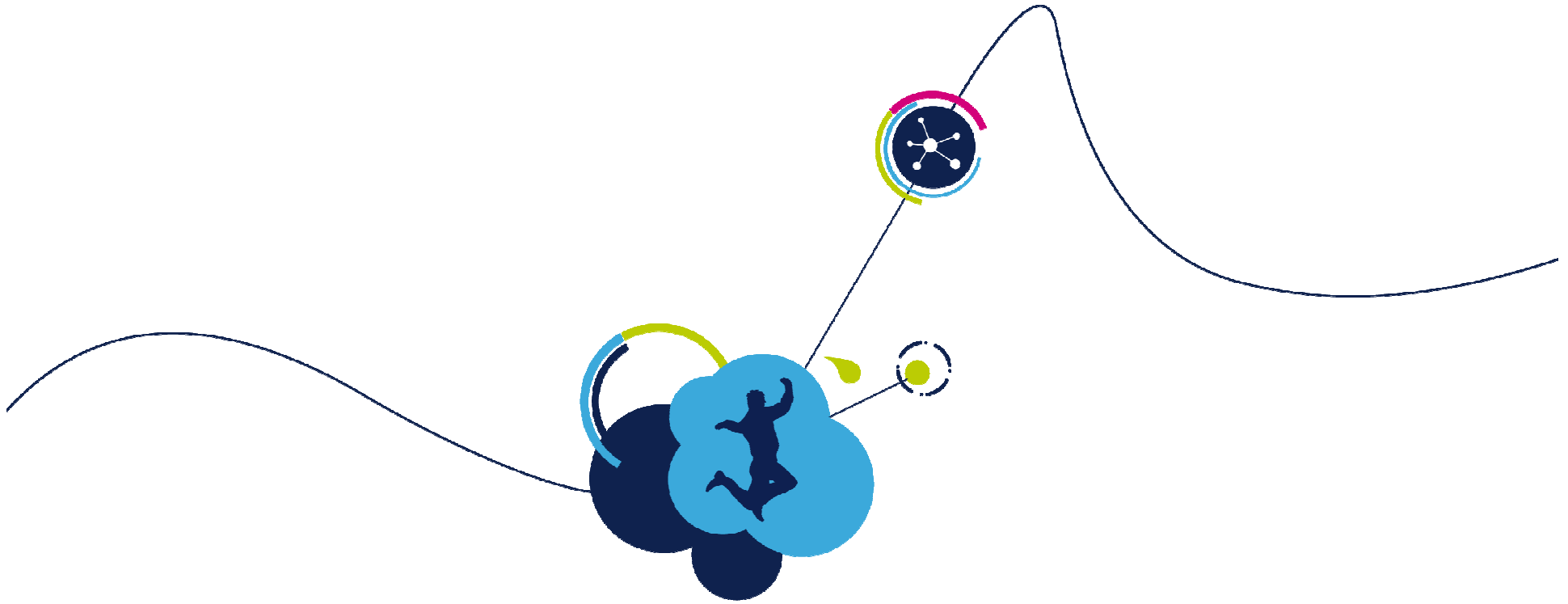
System blocks

General Purpose Input Output (GPIO)

General-Purpose I/Os (GPIO)

11

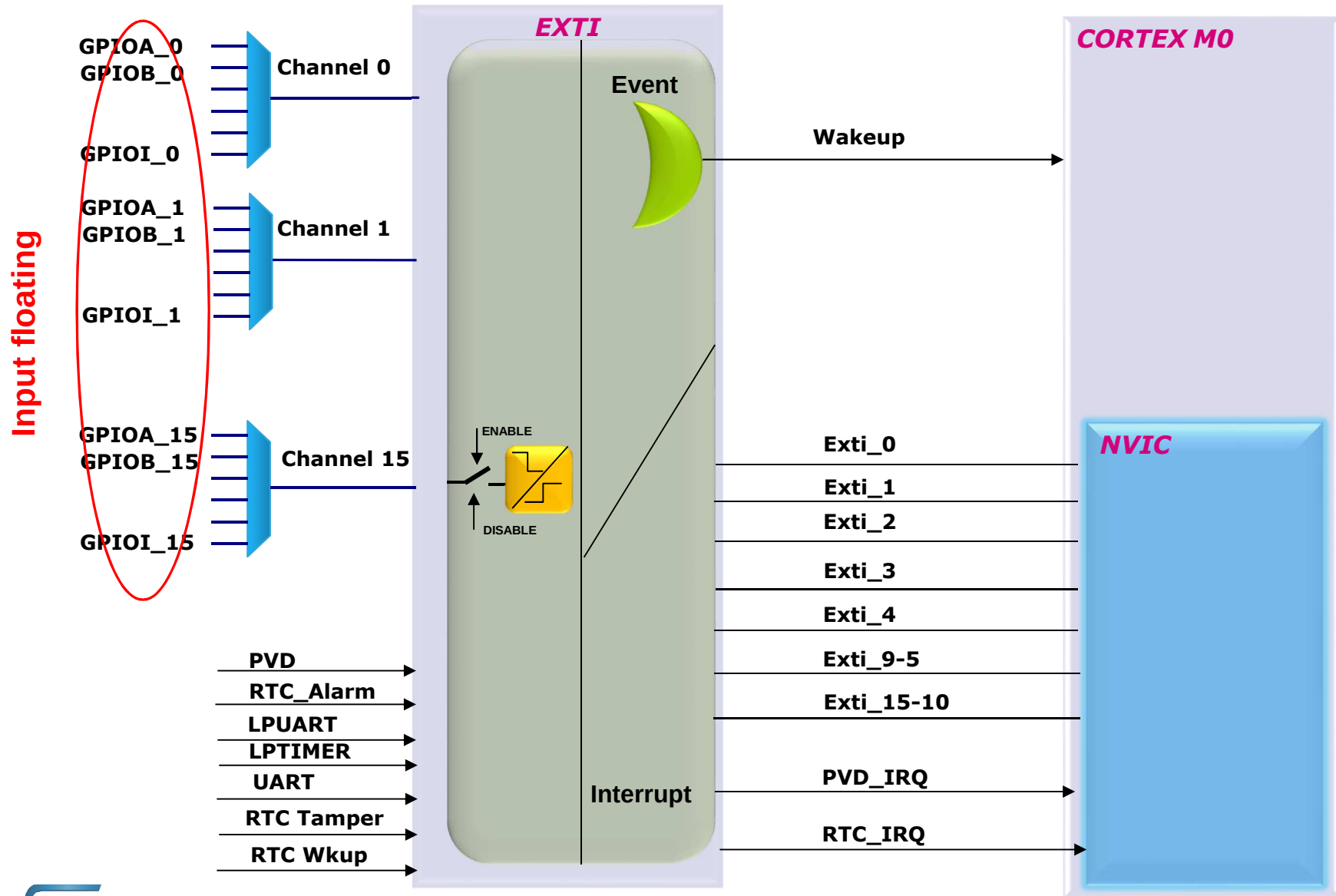


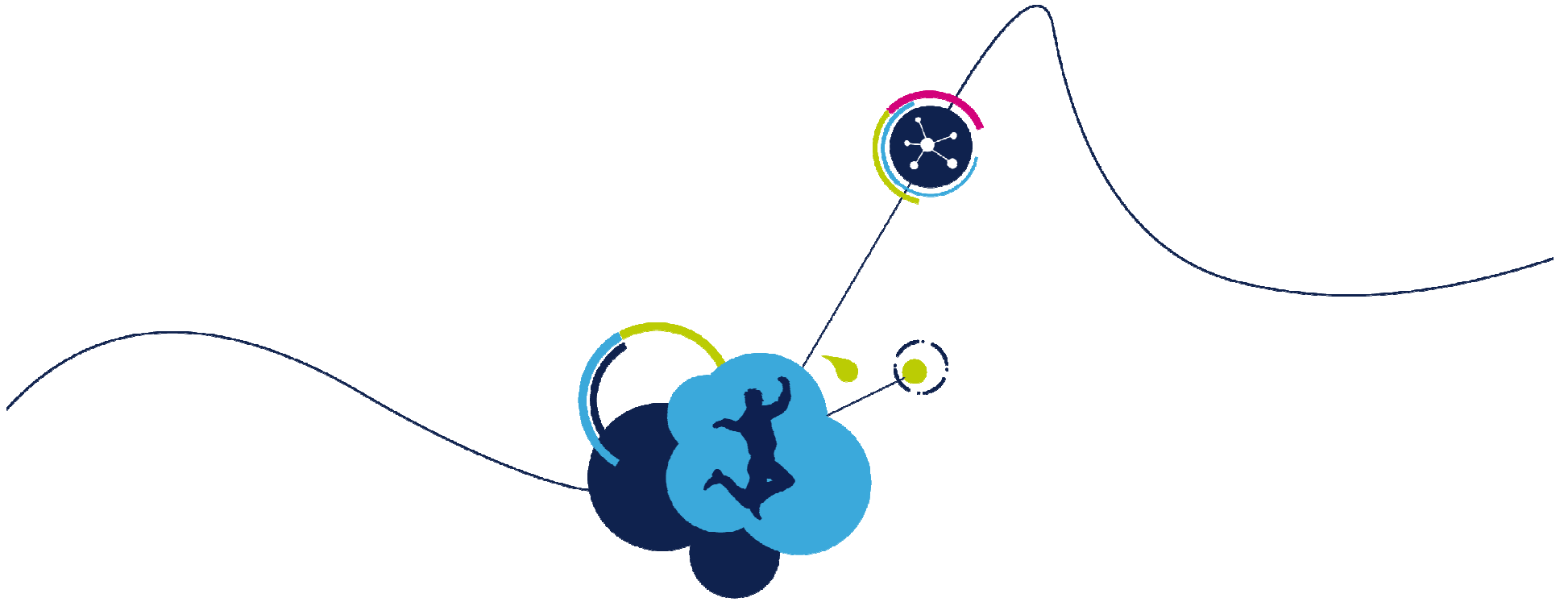


System blocks

Extended Interrupts/Events Controller (EXTI)

EXTI module: from pin to NVIC





Low-Power Modes

STM32L0 implementation

System Operation Modes

RUN

LP RUN = V_{Core} domain 131kHz max

SLEEP = CPU stopped

LP SLEEP = CPU stopped, V_{Core} domain 131kHz max

STOP = V_{Core} domain stopped, SRAM retained

STANDBY = V_{Core} domain switched OFF

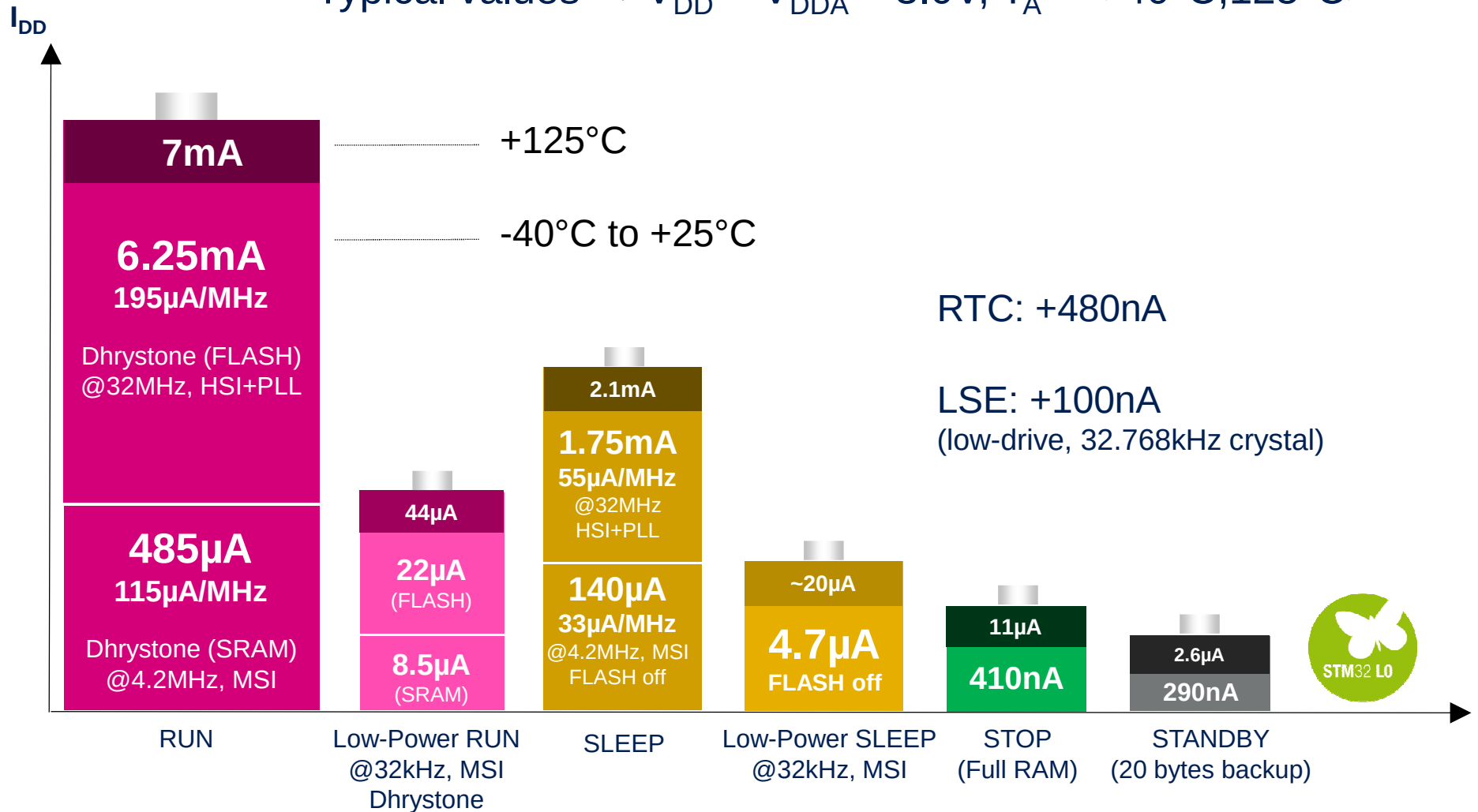
RUN

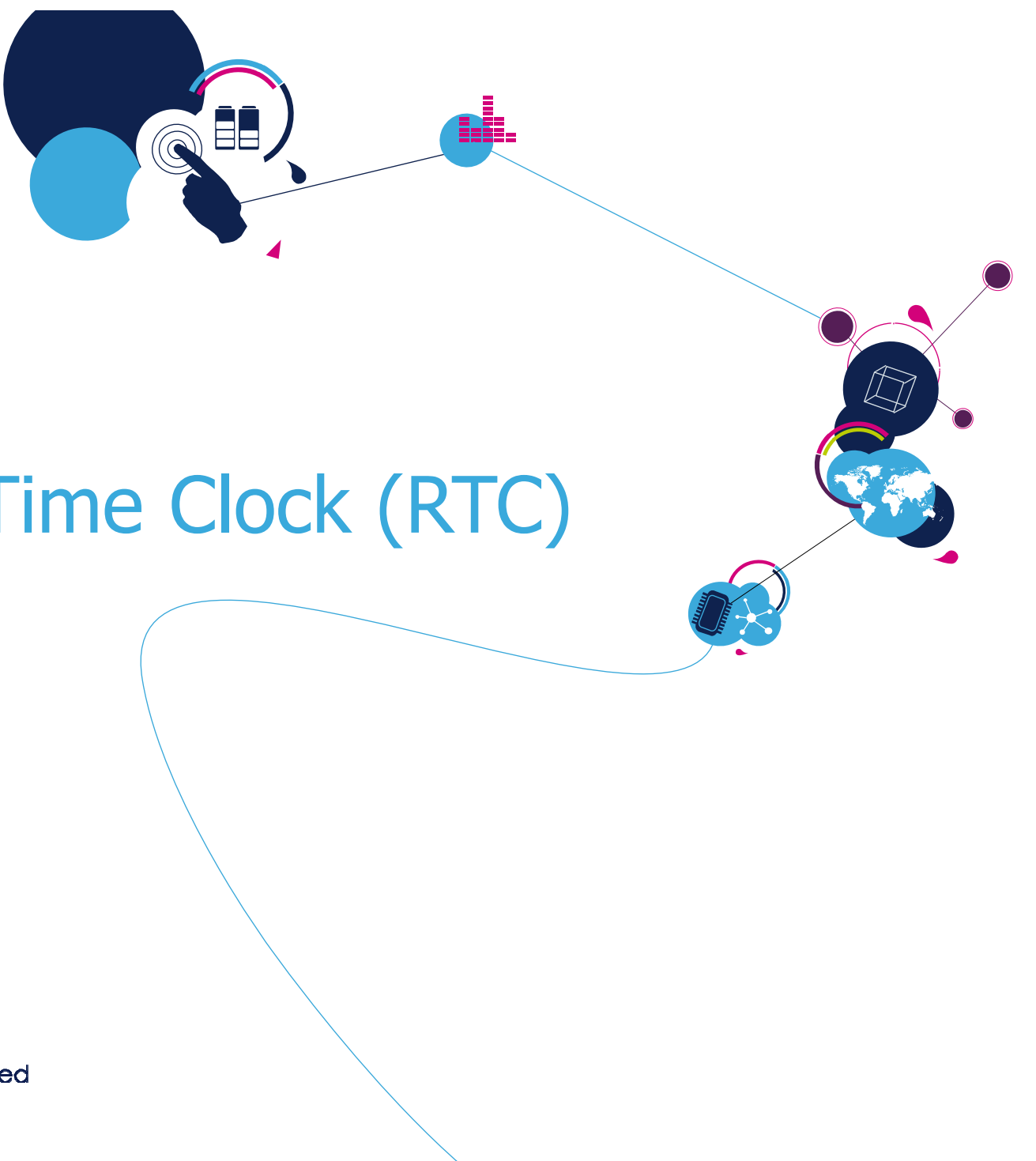
SLEEP

DEEP
SLEEP

STM32L05x - power consumption

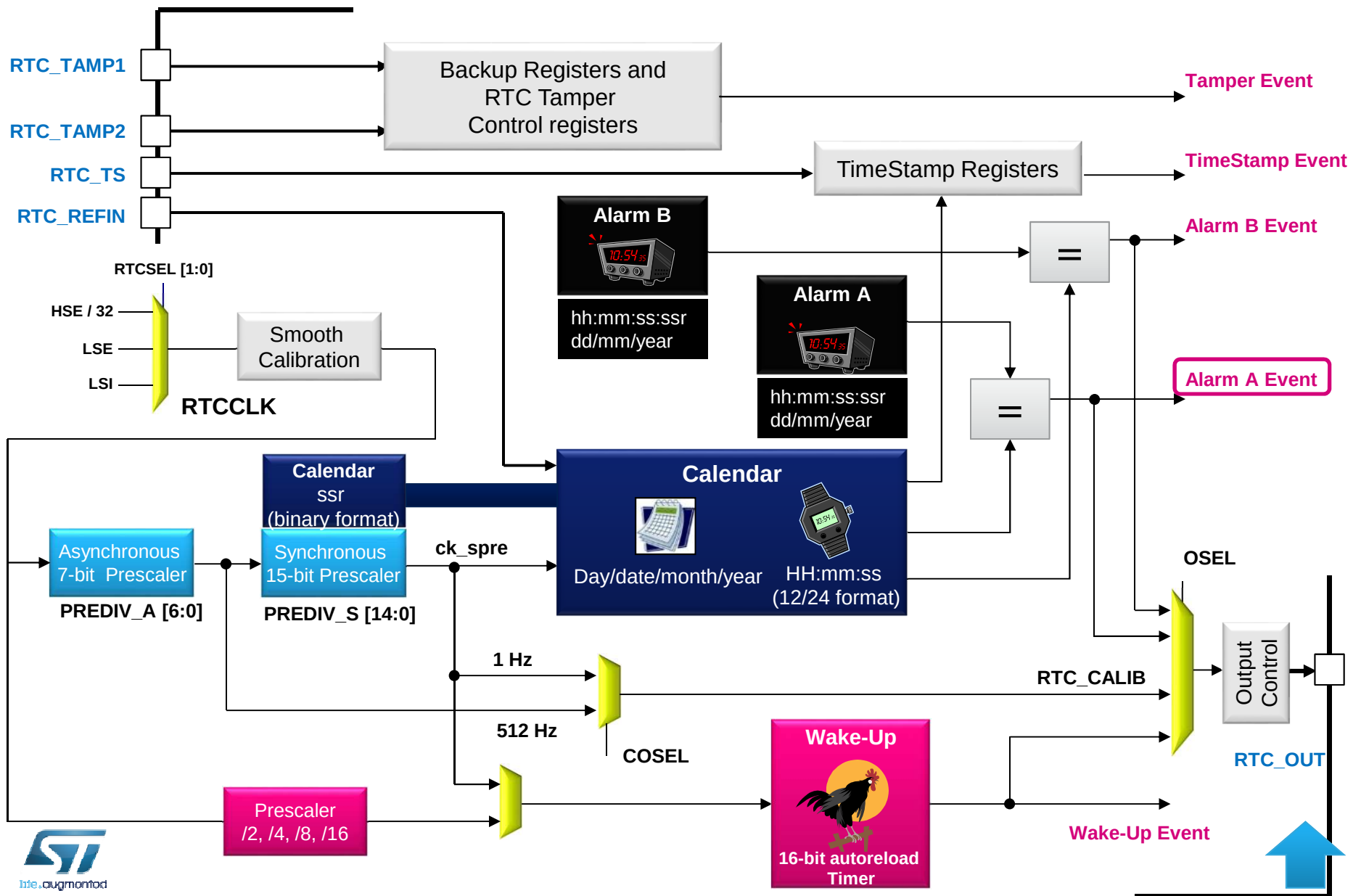
Typical values $\rightarrow V_{DD} = V_{DDA} = 3.0V$, $T_A = <-40^{\circ}C; 125^{\circ}C>$

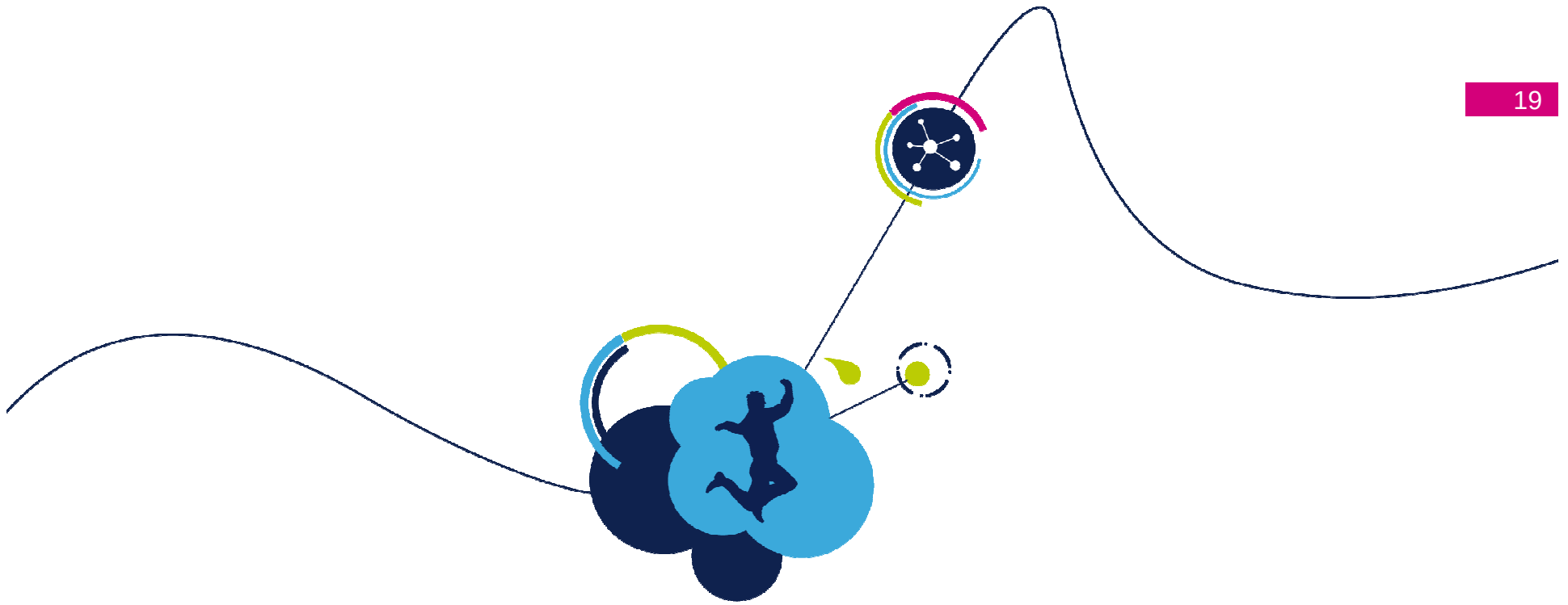




Real-Time Clock (RTC)

Block Diagram (RTC)





Digital peripherals

USART

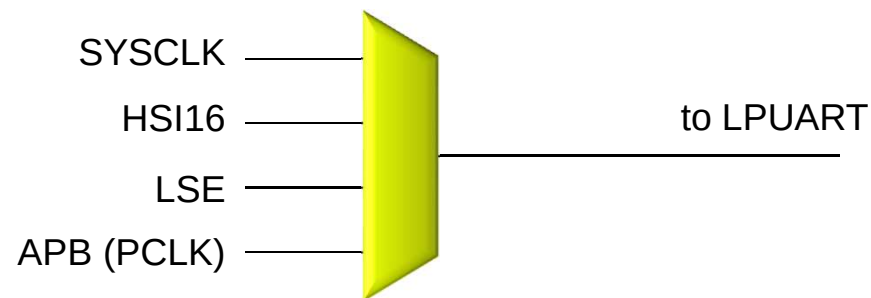
Frame	• 7, 8, 9 DATA bits • 0.5, 1, 1.5, 2 STOP bits • Even, odd, none PARITY • Oversampling /8 and /16 (default)	2x USARTs + 1x LPUART
Modes	• Asynchronous ▪ LIN ▪ SmartCard (T=0, T=1) ▪ IrDA SIR ENDEC ▪ Multiprocessor communication ▪ Half duplex ▪ Basic MODBUS • Synchronous (CLK line)	
Other	• DMA support • HW flow control (RTS, CTS lines) • Programmable data order (MSB/LSB) • Wake-Up from STOP mode • Swappable Tx/Rx pin, Driver Enable (for RS-485)	 



Digital peripherals

Low-Power UART (LPUART)

- LPUART includes all necessary hardware support to make asynchronous serial communications possible with minimum power consumption
- With just the **LSE 32.768 kHz** it is possible to run at up to **9600 baud**
- → For this purpose, the baudrate generation has been changed comparing to the USART peripheral

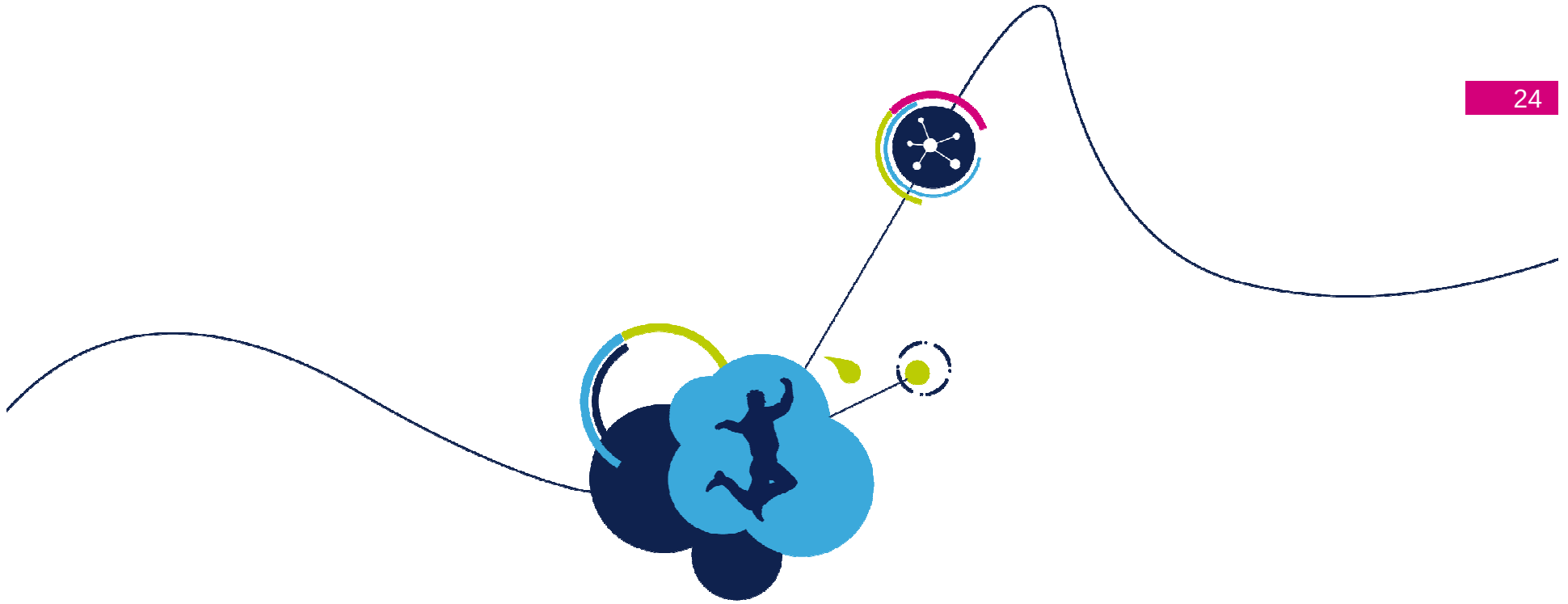


- Can be kept enabled and clocked even during STOP mode

USART/LPUART feature list

23

Features	USART1/2	LPUART1
Programmable data word length (7,8 or 9 bits)	•	•
Configurable stop bits (1 or 2)	•	•
Hardware Flow Control (Modem, RS-485 transceiver)	•	•
Continuous communication using DMA	•	•
Multi-processor communication	•	•
Single wire half duplex mode	•	•
Dual clock domain and Wake-Up from STOP mode	•	•
Swappable Rx/Tx pin configuration	•	•
Synchronous mode	•	
Smartcard mode	•	
IrDA	•	
LIN	•	
Receiver timeout	•	
Modbus Communication	•	
Autobaudrate detection	•	



Digital peripherals

General Purpose Timers

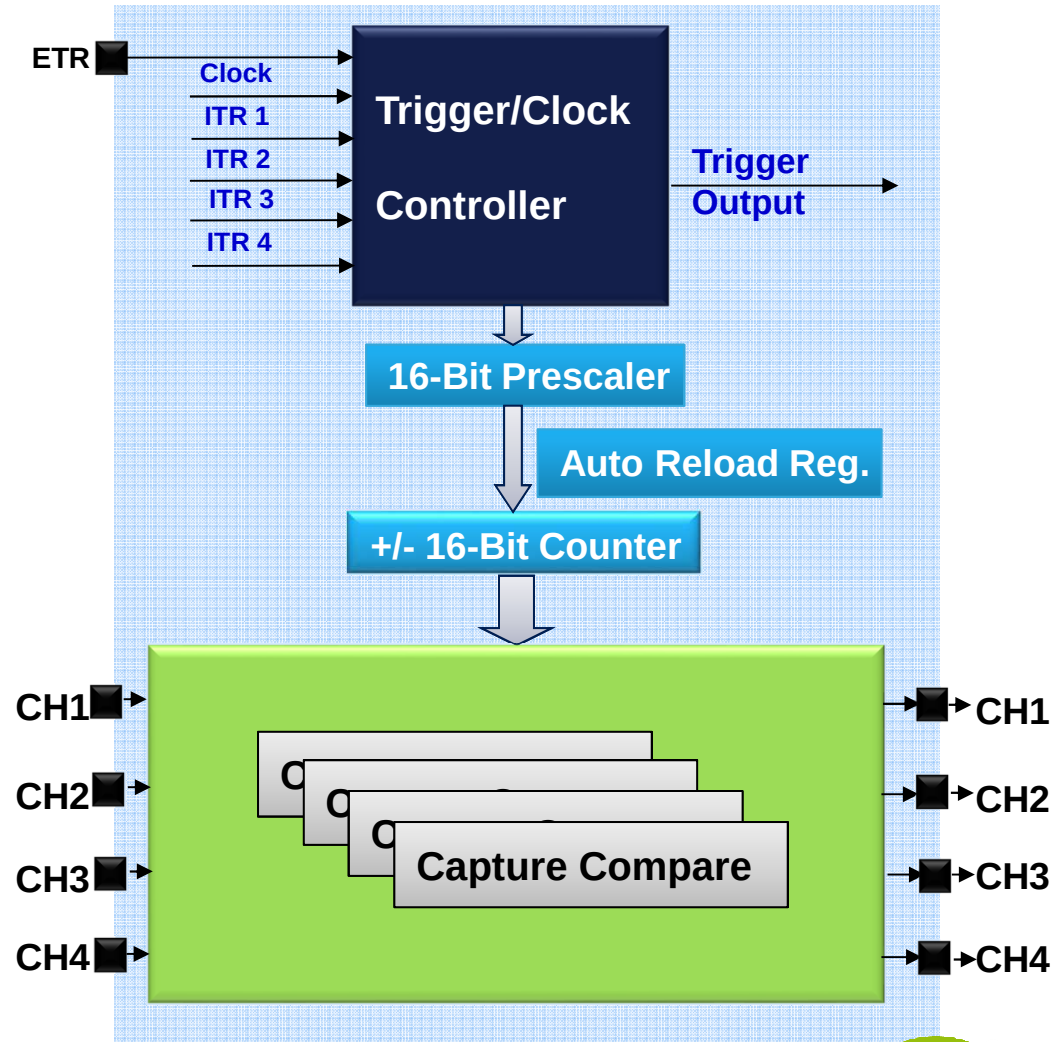
General Purpose Timers

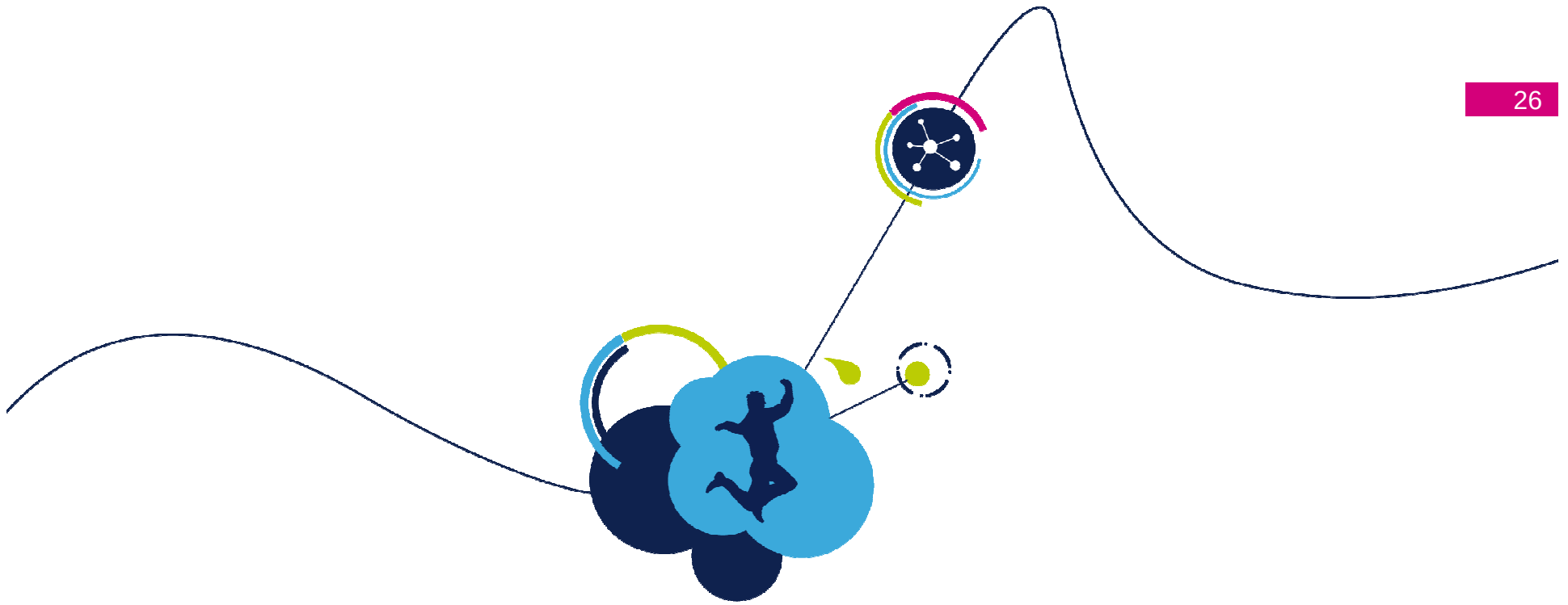
25

- **3 x 16-bit timer**
 - with autoreload
 - 8 CAPCOM units

Output Compare / Input Capture
PWM Output / Input
One Pulse Mode
Encoder interface
Synchronization
(Master/Slave, with external trigger)

- **1 x 16-bit basic timer**
 - with autoreload (DAC support)
- **1 x 16-bit Low-Power timer (LPTIM)**



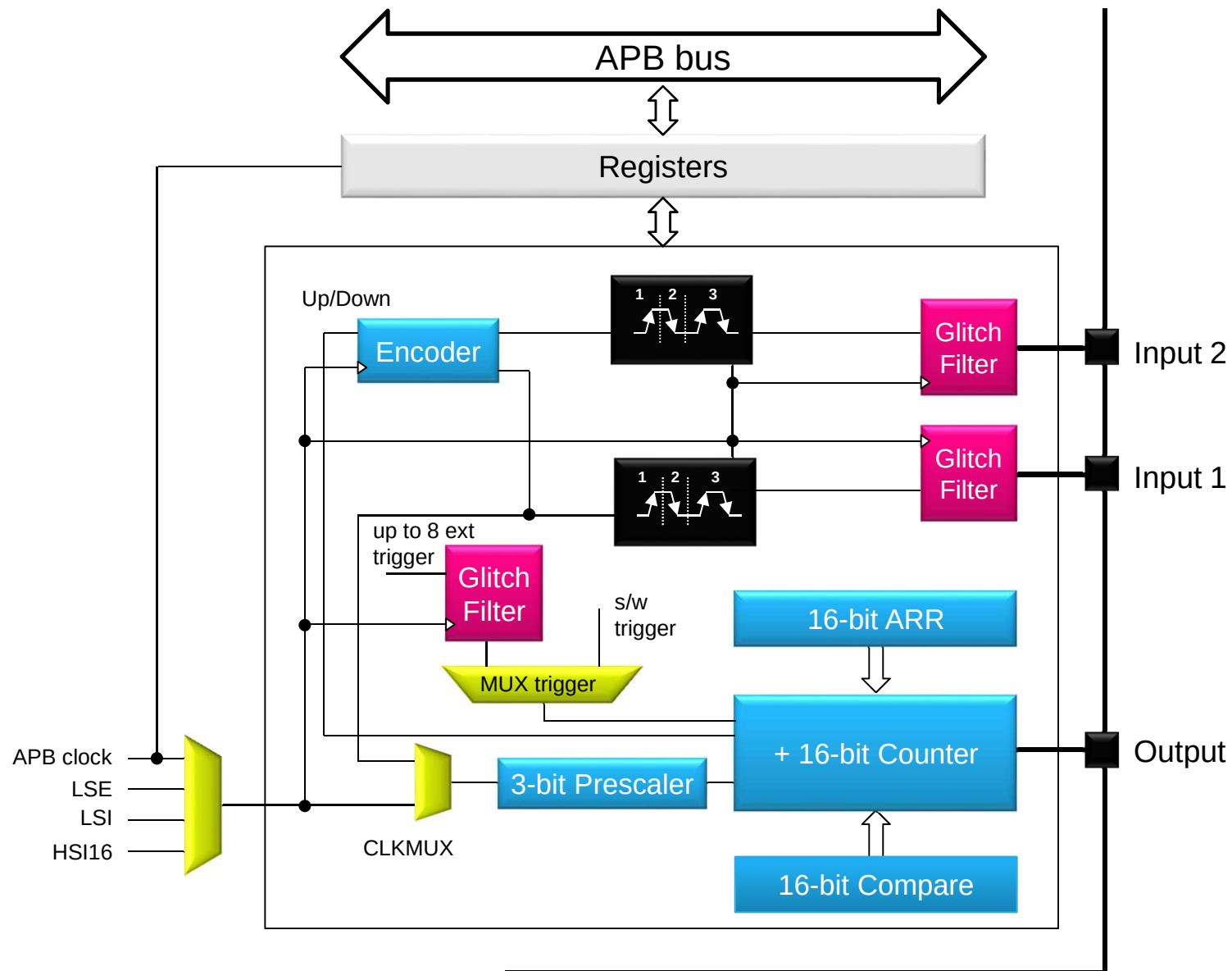


Digital peripherals

LPTIM

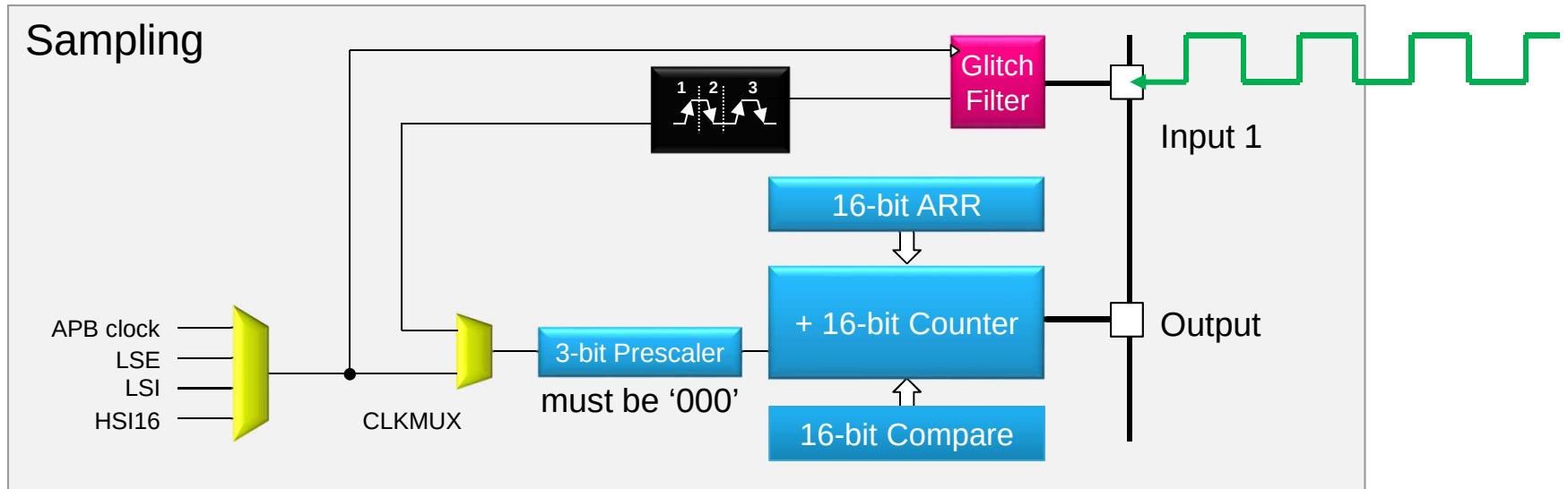
Low-Power Timer Block diagram

27

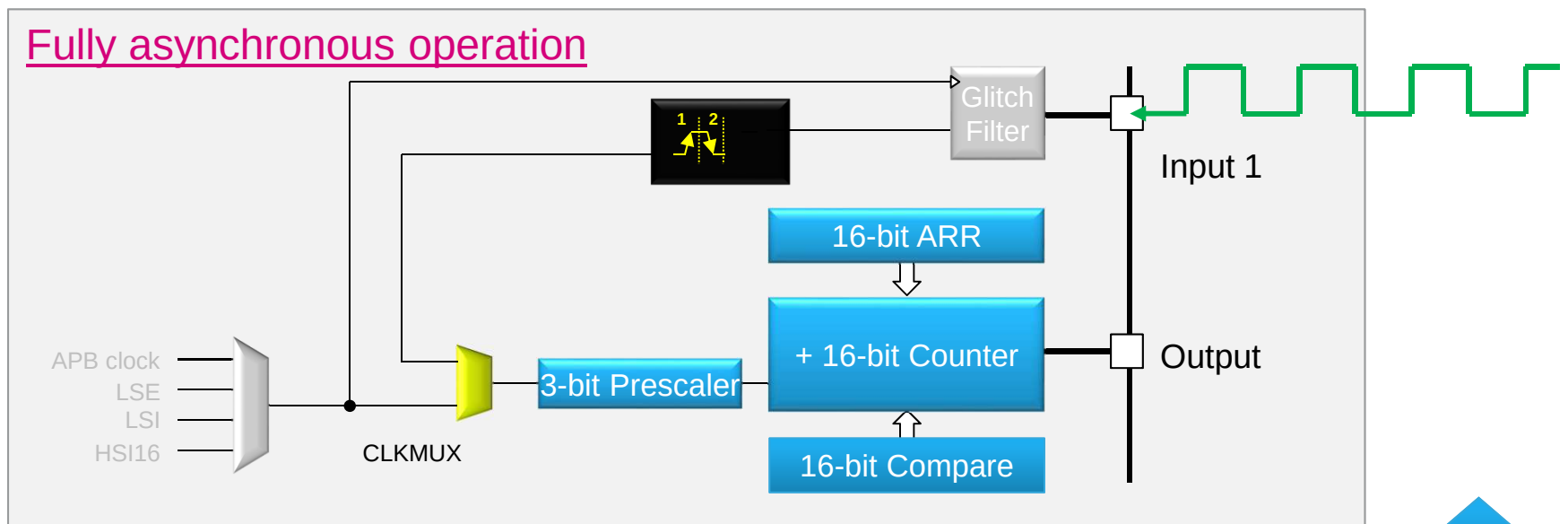


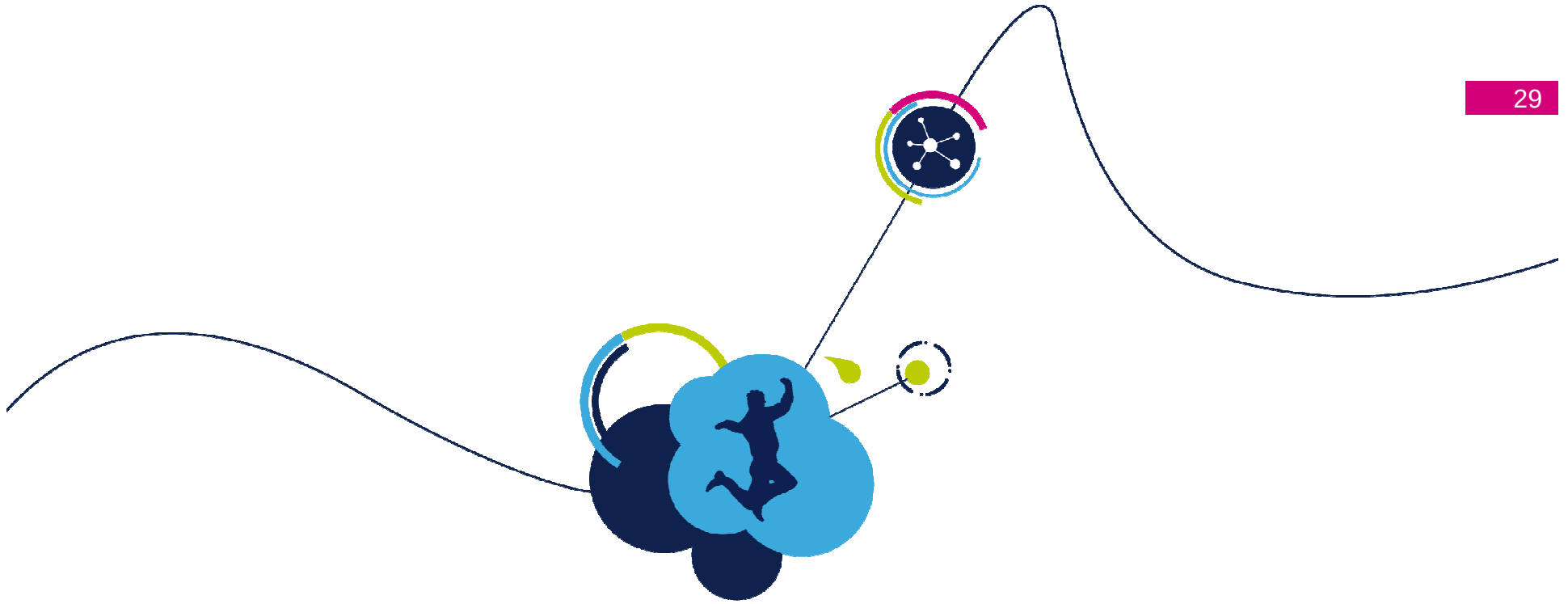


LPTIM as External Pulse Counter



Fully asynchronous operation





Analog peripherals

ADC

Analog to Digital Converter (ADC)

30

- ADC conversion rate 1.14 MSPS and 12-bit resolution (**0.87us @16MHz**)
 - **Up to 16-bit resolution with internal HW oversampler**
- Available in Performance and Low-power Run with CPU clk @ 32kHz
- ADC supply requirement: 1.65V to 3.6 V (from 2.4V full speed available)
- Up to **19 multiplexed channels** (16 external + 3 internal)
- Single and continuous conversion modes
- **Programmable sampling time**
- Hardware Delay insertion between conversions
- Programmable Conversion resolution : **12**, 10, 8 or **6-bit**
- **Analog Watchdog** on high and low thresholds

DMA

